The ClinGen Evidence Repository is an **FDA-recognized human genetic variant database** containing expert-curated assertions regarding variants' pathogenicity and supporting evidence summaries. [Disclaimer]

Variant: NM_000156.6(GAMT):c.268G>A (p.Glu90Lys)

Version: 1.0

CA402996483

916122 (ClinVar)

Gene: GAMT (HGNC:2593)
Condition: guanidinoacetate methyltransferase deficiency (MONDO:0012999)
Inheritance Mode: Autosomal recessive inheritance
UUID: 48f2aa0e-c95d-486e-a0b6-f29bce515120
Approved on: 2023-02-03
Published on: 2023-03-09

HGVS expressions

NM_000156.6:c.268G>A
NM_000156.6(GAMT):c.268G>A (p.Glu90Lys)
NC_000019.10:g.1399852C>T
CM000681.2:g.1399852C>T
NC_000019.9:g.1399851C>T
CM000681.1:g.1399851C>T
NC_000019.8:g.1350851C>T
NG_009785.1:g.6702G>A
ENST00000252288.8:c.268G>A
ENST00000447102.8:c.268G>A
ENST00000640762.1:c.199G>A
ENST00000252288.6:c.268G>A
ENST00000447102.7:c.268G>A
NM_000156.5:c.268G>A
NM_138924.2:c.268G>A
NM_138924.3:c.268G>A

Uncertain Significance

Met criteria codes 4

PP3 **PM3_Supporting** **PP4_Moderate** **PM2_Supporting**

Evidence Links 0

Expert Panel

Cerebral Creatine Deficiency Syndromes VCEP

Criteria Specification Information

- **Criteria Specification:** *ClinGen Cerebral Creatine Deficiency Syndromes Expert Panel Specifications to the ACMG/AMP Variant Interpretation Guidelines for GAMT Version 1.1.0*
- **Criteria Specification Approval History**
- **Criteria Specifications for this VCEP**

Evidence submitted by expert panel

Cerebral Creatine Deficiency Syndromes VCEP

The NM_000156.6:c.268G>A variant in GAMT is a missense variant that is predicted to result in the substitution of glutamate by lysine at amino acid 90 (p.Glu90Lys). One proband who is homozygous for the variant has been reported. This individual has development delay and seizures, in addition to elevated guanidinoacetate in urine and plasma and low creatine in urine and plasma (PMID: 24440240). The computational predictor REVEL gives a score of 0.943 which is above the threshold of 0.75, evidence that correlates with impact to GAMT function (PP3). To our knowledge, the results of functional studies are not available. The highest population minor allele frequency in gnomAD v2.1.1 is 0.000036 (1/27754 alleles) in the South Asian population, which is lower than the ClinGen CCDS VCEP’s threshold for PM2_Supporting (<0.0004), meeting this criterion (PM2_Supporting). The variant is noted in ClinVar (Variation ID: 916122). In summary, this variant meets the criteria to be classified as a variant of unceratinsignificance for GAMT deficiency. GAMT-specific ACMG/AMP codes met, as specified by the ClinGen Cerebral Creatine Deficiency Syndromes VCEP (Specifications Version 1.1.0): PP4_Moderate, PP3, PM3_Supporting, PM2_Supporting.

Met criteria codes

PP3 ✔	The computational predictor REVEL gives a score of 0.943 which is above the threshold of 0.75, evidence that correlates with impact to GAMT function (PP3).
PM3_Supporting ✔	One proband with clinical and biochemical features consistent with GAMT deficiency is homozygous for the variant. 0.5 points, PM3_Supporting.
PP4_Moderate ✔	One proband who is homozygous for the variant has been reported. This individual has development delay and seizures, in addition to elevated guanidinoacetate in urine and plasma and low creatine in urine and plasma (PMID: 24440240).
PM2_Supporting ✔	The highest population minor allele frequency in gnomAD v2.1.1 is 0.000036 (1/27754 alleles) in the South Asian population, which is lower than the ClinGen CCDS VCEP’s threshold for PM2_Supporting (<0.0004), meeting this criterion (PM2_Supporting).

Curation History

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